Question 2

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Question 2

- Dickey and Fuller consider three differential-form to detect the presence of a unit root

$$(1) \qquad \Delta Y_t = \gamma Y_{t-1} + \sum_{j=1}^{p} \left(\delta_j \Delta Y_{t-j} \right) + e_t$$

$$(2) \qquad \Delta Y_t = \alpha + \gamma Y_{t-1} + \sum_{j=1}^{p} \left(\delta_j \Delta Y_{t-j} \right) + e_t$$

$$(3) \qquad \Delta Y_t = \alpha + \beta t + \gamma Y_{t-1} + \sum_{j=1}^{p} \left(\delta_j \Delta Y_{t-j} \right) + e_t$$

Question 2

- The focus of testing is whether the coefficient γ is equal to zero, where $\gamma = 1 - \alpha_1 - \alpha_2 \ldots\ldots \alpha_p$, which means that the original process has a unit root.
- The null hypothesis of γ = 0, is tested against the alternative hypothesis γ < 0 of stationary.

Question 2

Unit root test on Eggs

Model with lag, constant and trend

L	ADF test statistic	Critical Value 1%	5%	10%
1	-1.988	-4.099	-3.477	-3.166
2	-2.186	-4.102	-3.478	-3.167
3	-2.196	-4.104	-3.479	-3.167
4	-2.359	-4.106	-3.480	-3.168

Model with lag and constant, but not trend:

L	ADF test statistic	Critical Value 1%	5%	10%
1	-0.809	-3.548	-2.912	-2.591
2	-1.167	-3.549	-2.912	-2.591
3	-1.205	-3.551	-2.913	-2.592
4	-1.423	-3.552	-2.914	-2.592

Model with lag, but no constant nor trend:

L	ADF test statistic	Critical Value 1%	5%	10%
1	1.719	-2.611	-1.95	-1.61
2	1.830	-2.611	-1.95	-1.61
3	1.844	-2.612	-1.95	-1.61
4	1.912	-2.612	-1.95	-1.61

Unit root test on Chickens

Model with lag, constant and trend

L	ADF test statistic	Critical Value 1%	5%	10%
1	-1.678	-4.099	-3.477	-3.166
2	-2.281	-4.102	-3.478	-3.167
3	-2.268	-4.104	-3.479	-3.167
4	-2.843	-4.106	-3.48	-3.168

Model with lag and constant, but not trend:

L	ADF test statistic	Critical Value 1%	5%	10%
1	-1.908	-3.548	-2.912	-2.591
2	-2.289	-3.549	-2.912	-2.591
3	-2.319	-3.551	-2.913	-2.592
4	-2.673	-3.552	-2.914	-2.592

Model with lag, but no constant nor trend:

L	ADF test statistic	Critical Value 1%	5%	10%
1	-0.185	-2.611	-1.95	-1.61
2	-0.100	-2.611	-1.95	-1.61
3	-0.155	-2.612	-1.95	-1.61
4	-0.111	-2.612	-1.95	-1.61

Question 2

- The unit root hypothesis is not rejected even at the 10 percent level.
- Regressions with nonstationary variables are spurious unless they are cointegrated, then the significance of the variables eggs in the regression of causality could be spurious.

Question 2

Granger Causality Tests.

Part 1: Did the Chicken Come First?

$$\Delta Eggs_t = \sum_{i=1}^{L} \alpha_i \, \Delta Eggs_{t-i} = \sum_{i=1}^{L} \beta_i \, \Delta Chikens_{t-i} + \epsilon_t$$

L = no. of lags	F statistic	P - value	R of the regression
1	0.1207	0.7294	0.1478
2	0.2472	0.7817	0.1479
3	0.1461	0.9318	0.1413
4	0.1995	0.9377	0.1432

Part 2: Did the Egg come first?

$$\Delta Chickens_t = \sum_{i=1}^{L} \alpha_i \, \Delta Chikens_{t-i} = \sum_{i=1}^{L} \beta_i \, \Delta Eggs_{t-i} + \epsilon_t$$

1	12.379	0.001	0.170
2	4.424	0.016	0.175
3	3.364	0.024	0.196
4	4.161	0.005	0.279

Question 2

- We still find that eggs Granger cause chickens.
- Rudebusch (1993) shows that U,S data on real GNP, which fails to reject the unit root hypothesis, also fails to reject a stationarity hypothesis when the latter is set up as the null.